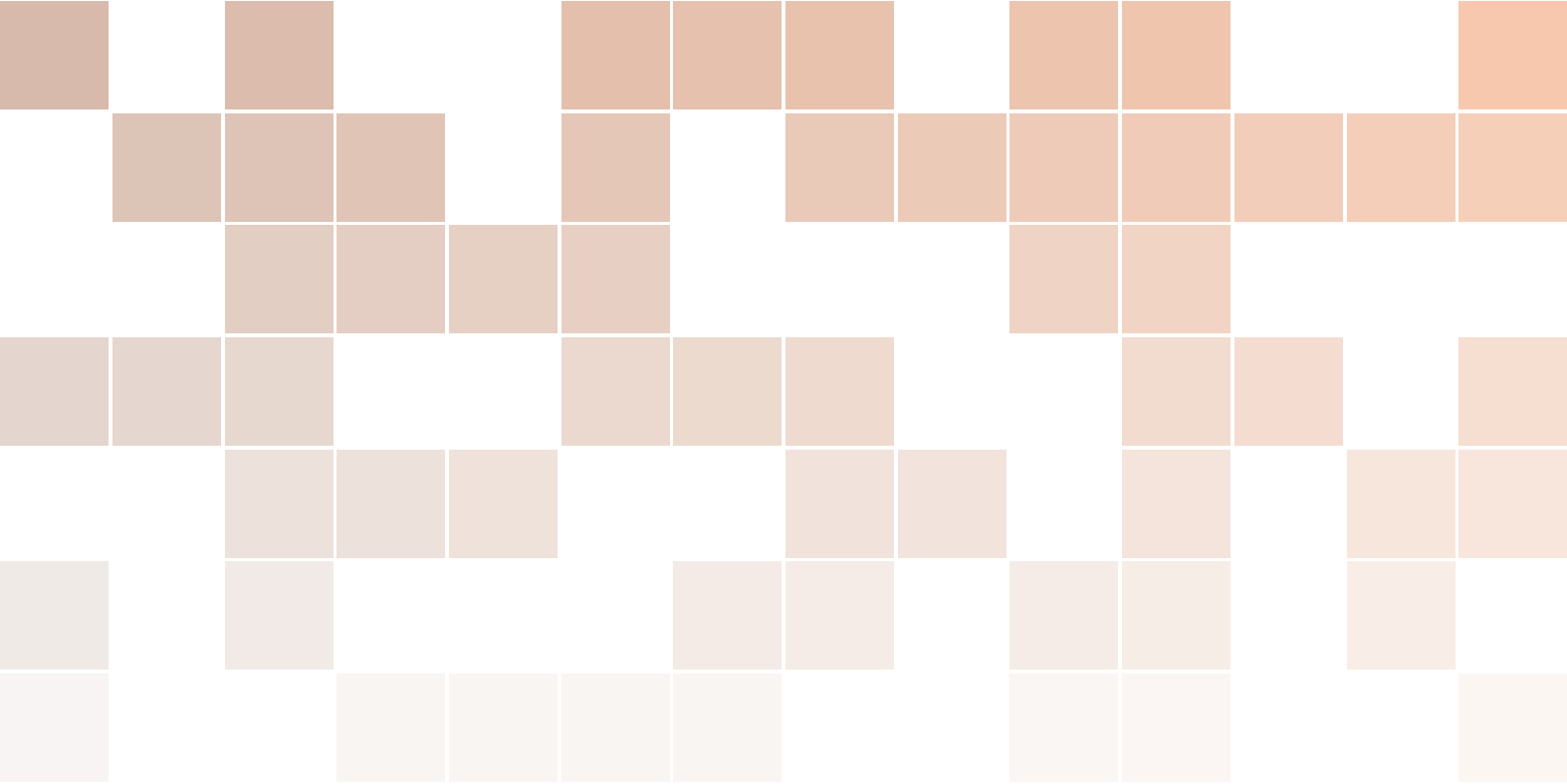
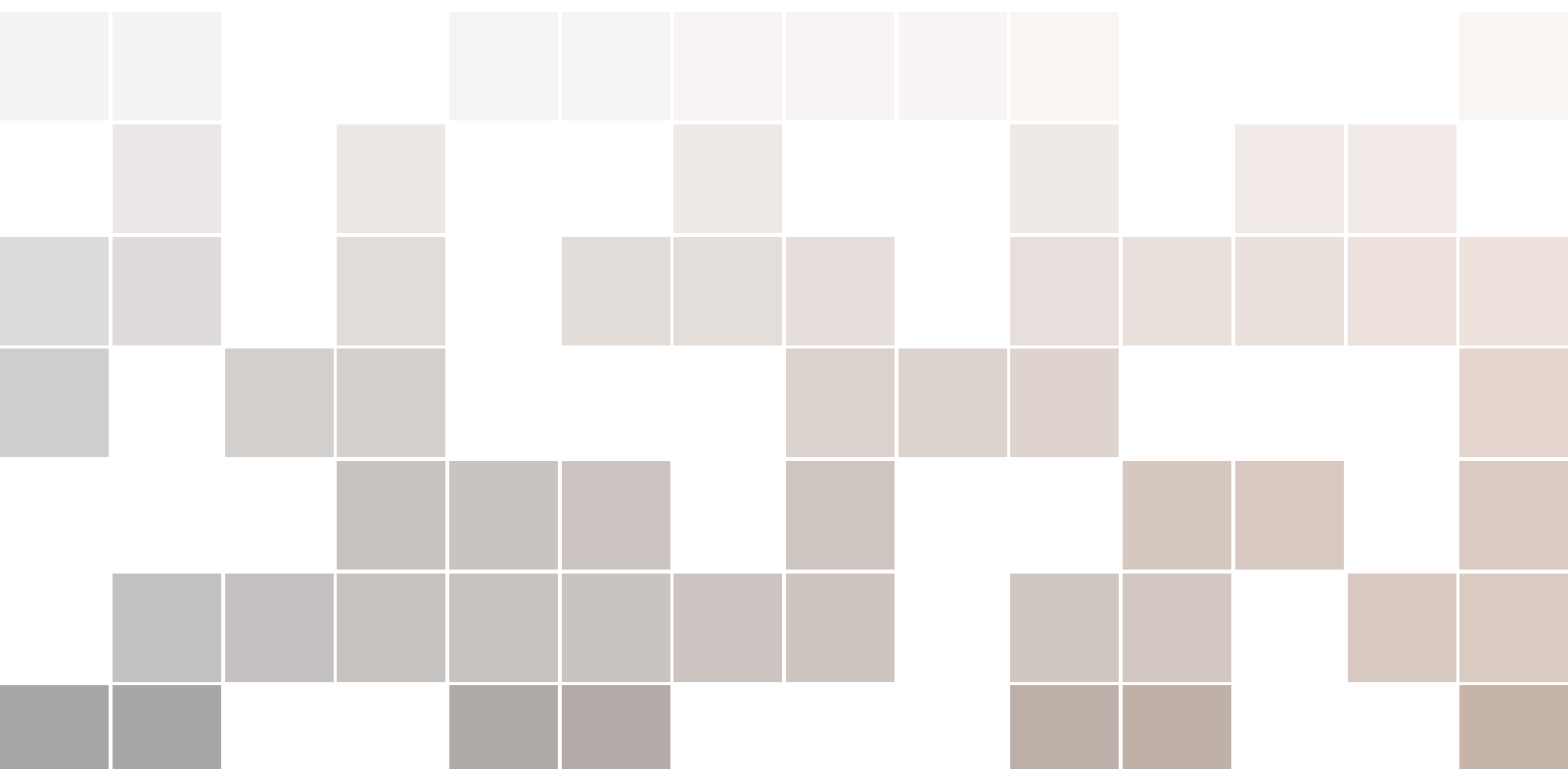




Lecture Notes on Quantum Computing

The first steps

Daniel Stilck França



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Bilbao, June 2024

Contents

I	Part One	
1	History and basics of quantum computing	7
1.1	Context and scope	7
1.1.1	Organization and further references	7
1.1.2	What this course will not be about	7
1.1.3	Introduction and Motivation	8
1.1.4	The central dogma of this course	9
1.1.5	Why Quantum Computation and Information?	9
1.1.6	History of the field	10
1.1.7	Modeling a Quantum System:	11
1.1.8	Probability Theory for Beginners	12
1.1.9	Quantum Mechanical Systems:	13
1.1.10	Transformations in quantum mechanics	13
1.1.11	Combining quantum systems	14
1.1.12	Dirac notation	15
1.1.13	Quantum circuits	15
1.1.14	The No-Cloning Theorem	15
1.1.15	Quantum Teleportation	17
1.1.16	Quantum Queries	18
2	Lecture 2	21
2.1	Quantum Circuits	21
2.1.1	Classical Computation with Quantum Circuits	22
2.2	The Early Quantum Algorithms	24
2.3	The Deutsch-Josza Algorithm	26

Bibliography	29
Books	29
Articles	29
Index	31



Part One

1	History and basics of quantum computing	
	7	
1.1	Context and scope	
2	Lecture 2	21
2.1	Quantum Circuits	
2.2	The Early Quantum Algorithms	
2.3	The Deutsch-Josza Algorithm	
	Bibliography	29
	Books	
	Articles	
	Index	31

1. History and basics of quantum computing

1.1 Context and scope

These lecture notes are for a 4 lecture course on the basics of quantum computing taught in Bilbao in the summer of 2024. Its target audience are mathematicians not familiar with field of quantum computing and who want to have a quick overview of what the field is about. Quantum computing has developed to a mature and diverse field of study at the intersection of physics, mathematics and computer science. As such, it is impossible to cover even a fraction of it in the span of ten hours.

In these lectures, we will focus on the fundamentals of quantum mechanics from a quantum computing perspective and then go through the foundations of quantum signal processing. This choice of syllabus is motivated by the fact that many modern quantum algorithms can be formulated in the language of quantum signal processing and, as such, this should give a good foundation to the audience members interested in deepening their understanding of quantum computing.

1.1.1 Organization and further references

- Lectures: 2 hours every Tuesday from 10:15-12:15 with a 5-minute break after one hour.
- If you have any comments or questions, please send me an email.
- Reading materials and lecture notes will be provided on my website [here](#).
- We will mostly follow the lecture notes of de Wolf, which are available [here](#). But there are other great resources to learn more about quantum computing! The standard reference for quantum computing is [1], which is available [here](#). Another recent set of lecture notes I liked is the one by Richard Kueng, which you find [here](#). All these notes go significantly beyond the topics we will be able to cover and are a great place to start if you wish to go deeper into quantum computing.
- If you prefer to watch a longer lecture series instead, I recommend the courses by John Preskill, available [here](#). Another good option is the by O'Donnell, available [here](#).

1.1.2 What this course will not be about

Quantum computing is a very diverse subject. In four lectures we can at best touch the surface and we will not discuss some important aspects of the field. In particular, we will not discuss the

following fundamental aspects:

1. Physical implementations of quantum computers: we will not discuss how to actually *build* a quantum computer, the current proposals and the underlying physics.
2. Quantum error correction and fault-tolerance: quantum computers are inherently noisy and it is nontrivial how to perform quantum computations with access to imperfect operations. This is the focus of study of quantum error correction. Here, we will assume that we are given perfect, noiseless quantum computers.
3. Alternative models of quantum computation: there are various ways to model quantum computers (adiabatic, measurement-based...). Although all these models deserve attention, we will here focus on the gate-based model, which is the closest to classical computation.

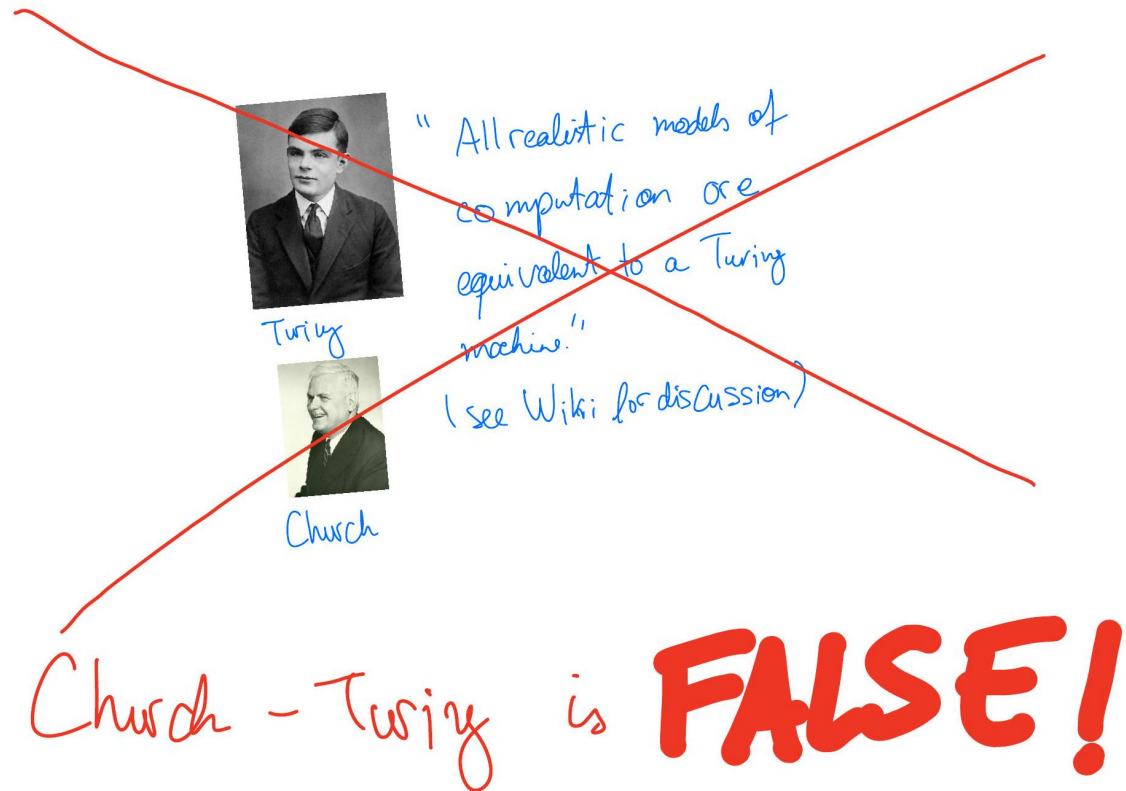
1.1.3 Introduction and Motivation

Classical computers (i.e. the phone in your pocket), regardless of their specific designs and implementations, fundamentally operate on the same principle: they use bits, which can take on the value of either 0 or 1, to represent variables. These bits are manipulated through programs, allowing the computer to perform a wide range of tasks. Over the years, we have observed a steady increase in both the speed at which these operations are carried out and the overall capability of computers. This has enabled more complex and computationally demanding tasks to be executed more efficiently.

Typically, improvements in computer hardware lead to constant reduction in runtime, and more sophisticated algorithms can lead to more substantial improvements. However, there are inherent limitations to these improvements. Specifically, if a problem is expected to fundamentally take exponential time to solve, no amount of hardware enhancement or algorithm optimization can reduce this exponential gap significantly. This inherent limitation brings us to the Strong Church-Turing thesis, which posits that all realistic models of computation are equivalent to a Turing machine in terms of their computational power.

The Strong Church-Turing thesis is a fundamental concept in computer science. It asserts that any physically realizable computational device can be simulated by a Turing machine with at most a polynomial time overhead. This implies that the class of problems solvable by such devices is essentially the same as those solvable by a Turing machine. In particular, if a problem requires exponential time to solve on a Turing machine, it must take exponential time to solve on any "reasonable" machine. We will discuss shortly what we mean here by reasonable.

1.1.4 The central dogma of this course



Our central dogma is that the Church-Turing hypothesis is **wrong**. Quantum computers can solve certain problems exponentially faster than classical machines! They are feasible and reasonable, though some critics, like Gil Kali (check his blog), may disagree.

1.1.5 Why Quantum Computation and Information?

If you do not care about the Church-Turing hypothesis or exponential speedups, another reason why you might think quantum computing is cool is because it involves some of the most exciting scientific breakthroughs of the the 20th century: quantum mechanics, computing and information theory. Thus, it is a grea field to do cool math!

→ Planck

Quantum theory

computer science

→ Turing

information theory

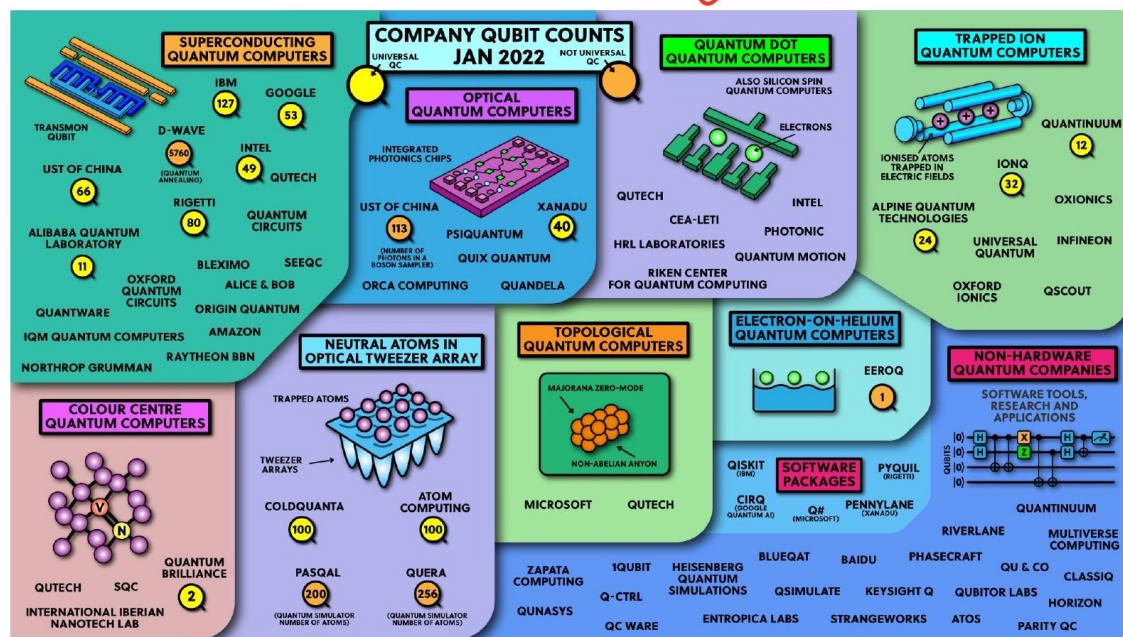
→ Shannon

. unites some of the coolest ideas of 20th century!

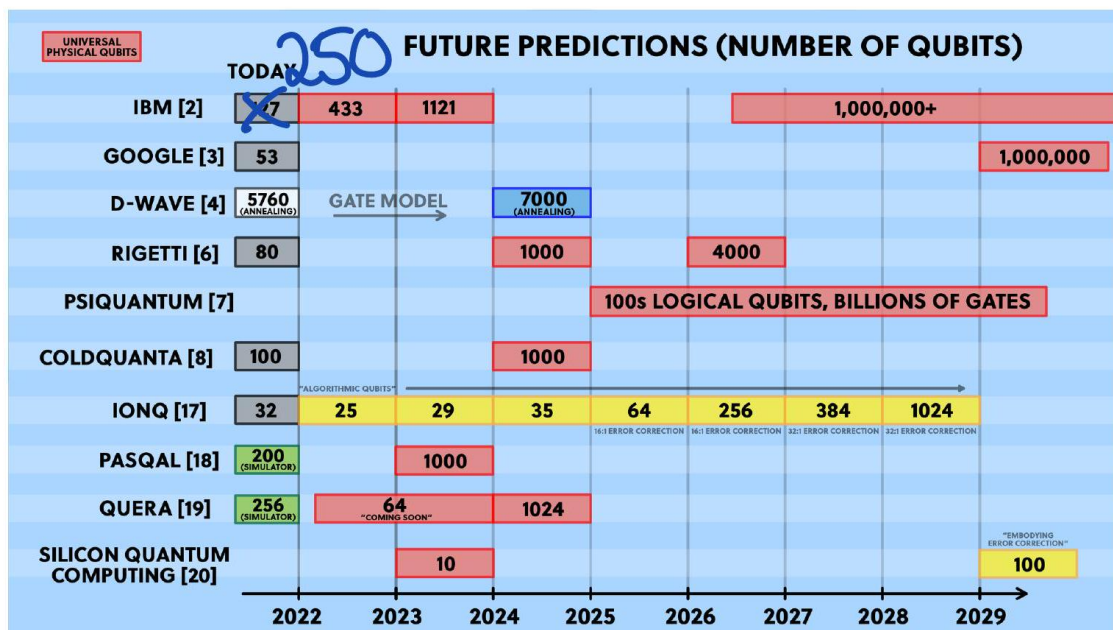
1.1.6 History of the field

Here is a short timeline of quantum computing:

- **80s:** Manin and Feynman's ideas of quantum mechanical models of computation.
- **1994:** Shor's factoring algorithm: Quantum computers can factor integers efficiently. This is not expected for classical computers. But what about the reasonable part of the Church-Turing thesis?
- **1995:** Shor: Quantum error correction is possible.
- **1996-1998:** Various threshold theorems by Aharonov, Kitaev, Laflamme, Gottesman, Preskill, and others. In essence, as long as noise is below a constant threshold, arbitrarily long computations are possible with an efficient overhead. These results form the core of why we believe quantum computing is reasonable!
- **1996:** Grover's algorithm. While Grover's algorithm provides only quadratic speedup, it is broadly applicable. See essays by Shor for further insights.



Graphics by Dominic Walliman on the current state of affairs. Already slightly outdated.



2021: First principles of error correction by Wallraff et al.

1.1.7 Modeling a Quantum System:

Quantum mechanics can be viewed as a generalization of probability theory. While probability theory deals with the likelihood of various outcomes in a given scenario, quantum mechanics extends this by incorporating the principles of wave functions and superposition. This allows for the description of phenomena that cannot be fully explained by classical probability theory alone, such as the behavior of particles at the quantum level where outcomes are not just uncertain but fundamentally probabilistic in a different manner.

In our exploration, we will build a connection between quantum mechanics and probability theory. This connection is crucial for understanding how quantum mechanics redefines our approach to probabilities. In classical probability theory, probabilities are real numbers between 0 and 1 that describe the likelihood of outcomes. In contrast, quantum mechanics uses complex probability amplitudes, which can interfere with each other, leading to phenomena like quantum superposition and entanglement. By establishing this connection, we aim to bridge the conceptual gap between classical and quantum perspectives on probability.

It is important to note that our approach to quantum mechanics is through the lens of quantum computing. This perspective emphasizes the computational aspects of quantum mechanics, focusing on how quantum systems can be harnessed for processing information in ways that classical computers cannot. As we delve into this subject, we will highlight how the principles of quantum mechanics are applied in the realm of quantum computing, offering a unique and powerful approach to computation. This course will not delve deep into Schrödinger's equations; we'll focus on finite-dimensional systems and will not touch upon topics typically discussed in an introductory quantum mechanics lecture like hydrogen atoms.

1.1.8 Probability Theory for Beginners

For a system with d states the state space can be described by $S = \{0, \dots, d-1\}$. For instance, $d = 2$ represents a coin (heads or tails).¹ In cases where we do not have perfect knowledge of outcomes, we assign a probability to each state. This assignment is illustrated by a probability vector.

$$\begin{bmatrix} p \\ 1-p \end{bmatrix} \begin{array}{l} \xrightarrow{\text{red}} 0/\text{heads state} \\ \xrightarrow{\text{blue}} 1/\text{tails state} \end{array}$$

These probabilities allow us to update our knowledge about the system. As for transformations, we assume we can manipulate such random variables and combine them.

Examples:

1. NOT gate: Flips 0 and 1, represented as

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

on a probability vector. Effect: exchange probabilities of 0 and 1.

2. CNOT: Acts on two input states. If the first state is 0, it does nothing. If the first state is 1, it flips the second state.

$$\begin{array}{ll} 00 \rightarrow 00 & 01 \rightarrow 01 \\ 10 \rightarrow 11 & 11 \rightarrow 10 \end{array}$$

Generally, any operation mapping a probability distribution is modeled by a stochastic matrix. We identify three primary components of a probabilistic theory:

- Identifying possible events.
- Assigning a probability to each event.
- Transforming the events.

¹Attention: even though I will try to be consistent, sometimes I will start counting from $i = 1$ and sometimes from $i = 0$. I apologize in advance if this happens.

1.1.9 Quantum Mechanical Systems:

Now that we discussed some fundamental concepts in probability theory, let us draw some parallels in quantum computation. We will use standard notions from linear algebra throughout the course.

Pure Quantum State: A pure quantum state of a d -level system is described by a unit vector $\psi \in \mathbb{C}^d$ such that $\sum_{i=0}^{d-1} |\psi_i|^2 = 1$. It describes the state of the system.

Measurement: In quantum mechanics we assign probabilities to different events with the help of measurements. A projective measurement on a d -level system is given by a set of orthogonal projections $\{P_j\}_{j=1}^m$ with $P_j \in \mathbb{C}^{d \times d}$ and $\sum_{j=1}^m P_j = I$. Each P_j corresponds to an outcome j .

Recall: A matrix P is called an orthogonal projection if:

1. $P^2 = P$
2. $P = P^\dagger$

Born's Rule (assigning probabilities): Given a state ψ and a measurement $\{P_j\}_{j=1}^m$, the probability of observing outcome j is given by

$$p(j) = \langle \psi | P_j | \psi \rangle = \|P_j \psi\|^2$$

Lemma 1.1.1 p is a valid probability distribution.

Proof. Let us check that both defining properties of a probability distribution (positivity and normalization) are satisfied:

1. $p(j) \geq 0$: $\langle \psi | P_j | \psi \rangle = \langle \psi | P_j^2 | \psi \rangle = \langle \psi | P_j^\dagger P_j | \psi \rangle = \|P_j \psi\|^2$
2. $\sum_j p(j) = \sum_j \langle \psi | P_j | \psi \rangle = \langle \psi | \sum_j P_j | \psi \rangle = \langle \psi | \psi \rangle = 1$.

■

R Note that quantum states are globally defined only up to a phase. For a projector P_j , we have

$$\langle e^{i\theta} \psi | P_j | e^{i\theta} \psi \rangle = e^{-i\theta} e^{i\theta} \langle \psi | P_j | \psi \rangle$$

indicating the same outcomes. Thus, quantum states are only defined up to a phase.

R For an explanation of why one would define a theory this way, please refer to "Gleason's theorem."

■ **Example 1.1 — Qubit.** A qubit is a 2-level system $\mathbb{C}^2$ given by

$$\psi = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \quad |\alpha|^2 + |\beta|^2 = 1$$

■ **Example 1.2 — Computational basis measurement.** For $\mathbb{C}^d$, the measurement is defined by the matrices $\{P_i\}_{i=1}^d$ where $(P_j)_{hl} = \delta_{(j,j),(h,l)}$.

1.1.10 Transformations in quantum mechanics

In quantum mechanics, transformations are described by unitary matrices.

Recall: A matrix $U \in \mathbb{C}^{d \times d}$ is called unitary if $U^\dagger U = U U^\dagger = I$.

Why? If U is unitary, we have $\|U\psi\| = \|\psi\|$ for all $\psi \in \mathbb{C}^d$, which means U preserves the state space and is invertible. The other direction also holds: if you have a transformation (not even necessarily linear) that preserves the scalar product between all vectors, then it must be unitary or anti-unitary. This is known as Wigner's theorem.

Some examples of unitary transformations:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$X \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} \beta \\ \alpha \end{pmatrix}$$

$$\text{ii) } H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$H \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\text{iii) } R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

1.1.11 Combining quantum systems

Composite systems are described by tensor products. Given $\varphi \in \mathbb{C}^{d_1}$ and $\psi \in \mathbb{C}^{d_2}$, we define the tensor product $\varphi \otimes \psi \in \mathbb{C}^{d_1 \cdot d_2}$ as

$$\varphi \otimes \psi = \begin{pmatrix} \varphi_1 \psi \\ \varphi_2 \psi \\ \vdots \\ \varphi_{d_1} \psi \end{pmatrix}$$

For tensor product of matrices: given $A \in \mathbb{C}^{d_1 \times d_1}$ and $B \in \mathbb{C}^{d_2 \times d_2}$ we define $A \otimes B \in \mathbb{C}^{d_1 d_2 \times d_1 d_2}$ as

$$A \otimes B = \begin{pmatrix} A_{11}B & A_{12}B & \cdots & A_{d_1 1}B \\ \vdots & \ddots & & \vdots \\ A_{d_1 1}B & A_{d_1 2}B & \cdots & A_{d_1 d_1}B \end{pmatrix}$$

If we have a system of n qubits, the state is described by a unit vector in $(\mathbb{C}^2)^{\otimes n}$. Example: CNOT gate action on 2 qubits is given by

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Why tensor products? the main reason we resort to tensor products is because they naturally reproduce the concept of independence. Recall that if we have two independent random variables X, Y , then for any two events A, B we have that:

$$\mathbb{P}(X \in A \cap Y \in B) = \mathbb{P}(X \in A) \mathbb{P}(Y \in B)$$

. We want the same to hold for independent/product quantum states. If we have two sets of orthogonal projections $\{P_i\}, \{Q_j\}$ defining two different measurements, we can define a new set of orthogonal projections $R_{i,j} = P_i \otimes Q_j$, which defines a new measurement with outcomes (i, j) . For two quantum states ψ_X, ϕ_Y we can define their joint state as $\psi_X \otimes \phi_Y$. Then, we have that the probability of measuring outcome (i, j) is given by:

$$\mathbb{P}(i, j) = \langle \psi_X \otimes \phi_Y | R_{i,j} | \psi_X \otimes \phi_Y \rangle = \langle \psi_X | P_i | \psi_X \rangle \langle \phi_Y | Q_j | \phi_Y \rangle.$$

Thus, we see that we recover the "product rule" for independent events.

How about disjoint events?

In the classical case, we have that if A, B are disjoint events, then $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$. In the quantum case, we have that if P, Q are orthogonal projections, then $P + Q$ is also an orthogonal projection. It is then not difficult to see that the probability of observing P or Q is given by:

$$\langle \psi | P + Q | \psi \rangle = \langle \psi | P | \psi \rangle + \langle \psi | Q | \psi \rangle.$$

This shows that we can model disjoint events with orthogonal projections. Note that the number of parameters required to describe an n -qubit system scales exponentially with n .

Example: CNOT gate action on 2 qubits is given by

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

1.1.12 Dirac notation

- A more convenient notation for matrix/vector manipulations used in Quantum Mechanics.
- We identify vectors $\psi \in \mathbb{C}^d$ with kets: $|\psi\rangle$.
- We identify elements of the canonical basis with integers:

$$e_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \equiv |0\rangle, \quad e_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \equiv |1\rangle$$

- We identify the conjugate transpose of vectors (often denoted by $\psi^\dagger$ or ψ^*) with bras: $\langle \psi |$.
- We write tensor products "next to each other". For example, $|0\rangle \otimes |0\rangle \equiv |00\rangle$.

1.1.13 Quantum circuits

Quantum circuits provide a simplified and useful way to represent quantum operations. These circuits are visualized using diagrams, which will be formally defined later. In these diagrams, time evolution is depicted as moving from left to right, making the temporal sequence of operations clear.

In a quantum circuit diagram:

1. Horizontal lines, or wires, represent different qubits.
2. Gates or operations are depicted as boxes placed on these wires.
3. Meters are used to denote measurements, indicating where and how qubits are measured in the circuit.

This graphical representation of quantum circuits helps in understanding and designing quantum algorithms by visually organizing the sequence and types of operations performed on qubits.

Let us discuss some examples:



Figure 1.1: Prepare the $|0\rangle$ state, applying a Hadamard gate, and measuring.

1.1.14 The No-Cloning Theorem

Now that we have established the basic notation of quantum mechanics, let us discuss one simple example where quantum information is different from classical information.

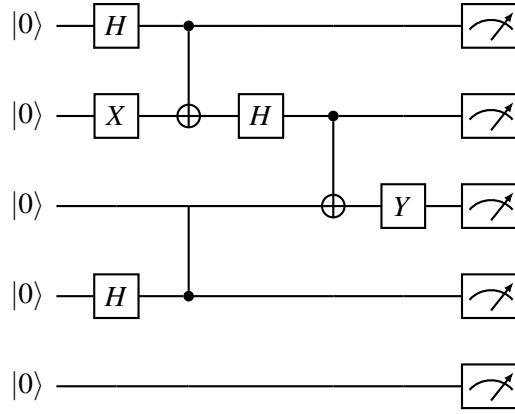


Figure 1.2: A more complex quantum circuit on 5 qubits. It has many controlled operations, which we will introduce later.

Theorem 1.1.2 — No-Cloning Theorem. There does not exist a cloning unitary $U \in \mathbb{C}^{d^2 \times d^2}$. That is, there is no unitary $U \in \mathbb{C}^{d^2 \times d^2}$ such that $\forall |\psi\rangle \in \mathbb{C}^d$ we have $U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle$.

Proof. Let's assume that there exists a unitary operator U that performs the cloning operation. Given two quantum states $|\psi\rangle$ and $|\phi\rangle$ that are orthonormal, the cloner should act as:

$$U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle$$

$$U|\phi\rangle|0\rangle = |\phi\rangle|\phi\rangle$$

where $|0\rangle$ is an auxiliary state used to store the clone.

Now, consider a superposition of the two states:

$$U\left(\frac{|\psi\rangle + |\phi\rangle}{\sqrt{2}}\right)|0\rangle$$

Using linearity of the unitary operation, we get:

$$U\left(\frac{|\psi\rangle + |\phi\rangle}{\sqrt{2}}\right)|0\rangle = \frac{U|\psi\rangle|0\rangle + U|\phi\rangle|0\rangle}{\sqrt{2}}$$

This gives:

$$\frac{|\psi\rangle|\psi\rangle + |\phi\rangle|\phi\rangle}{\sqrt{2}}$$

However, the correct cloning operation would yield:

$$\left(\frac{|\psi\rangle + |\phi\rangle}{\sqrt{2}}\right)\left(\frac{|\psi\rangle + |\phi\rangle}{\sqrt{2}}\right)$$

This is clearly not equal to the above result. Hence, we've reached a contradiction, implying that no such unitary operator U can exist. Thus, the no-cloning theorem is proven. ■



Figure 1.3: Source: xkcd

1.1.15 Quantum Teleportation

Quantum teleportation is a technique to transfer quantum information. This groundbreaking method was pioneered by Bennett et al. in 1993. The core task involves Alice wanting to send an unknown quantum state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ to Bob. Classically, transmission depends on precision. But, Alice does not even know α or β . Therefore, transmission is not possible classically.

Use of Entanglement

Additional resource - an entangled pair:

$$|\Omega\rangle = \frac{1}{\sqrt{2}}(|0\rangle_A|0\rangle_B + |1\rangle_A|1\rangle_B)$$

Alice holds one half, Bob the other. We will achieve quantum teleportation with 2 classical bits of communication and one shared maximally entangled state. The state $|\Omega\rangle$ is called maximally entangled.

Quantum Circuit for quantum teleportation

Let us now consider the quantum circuit for quantum teleportation described in Fig. 1.4. We will do a step-by-step analysis of the circuit.

State at step 1:

$$\begin{aligned} |\psi_1\rangle &= (\alpha|0\rangle + \beta|1\rangle) \left(\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle \right) \\ &= \frac{\alpha}{\sqrt{2}}|000\rangle + \frac{\alpha}{\sqrt{2}}|011\rangle + \frac{\beta}{\sqrt{2}}|100\rangle + \frac{\beta}{\sqrt{2}}|111\rangle \end{aligned}$$

State at step 2:

$$|\psi_2\rangle = \frac{1}{\sqrt{2}}(\alpha|000\rangle + \alpha|011\rangle + \beta|110\rangle + \beta|101\rangle)$$

State at step 3, recalling $H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$:

$$\begin{aligned} |\psi_3\rangle &= \frac{1}{2}(\alpha|000\rangle + \alpha|100\rangle + \alpha|011\rangle \\ &\quad + \beta|111\rangle + \beta|010\rangle - \beta|110\rangle - \beta|101\rangle) \end{aligned}$$

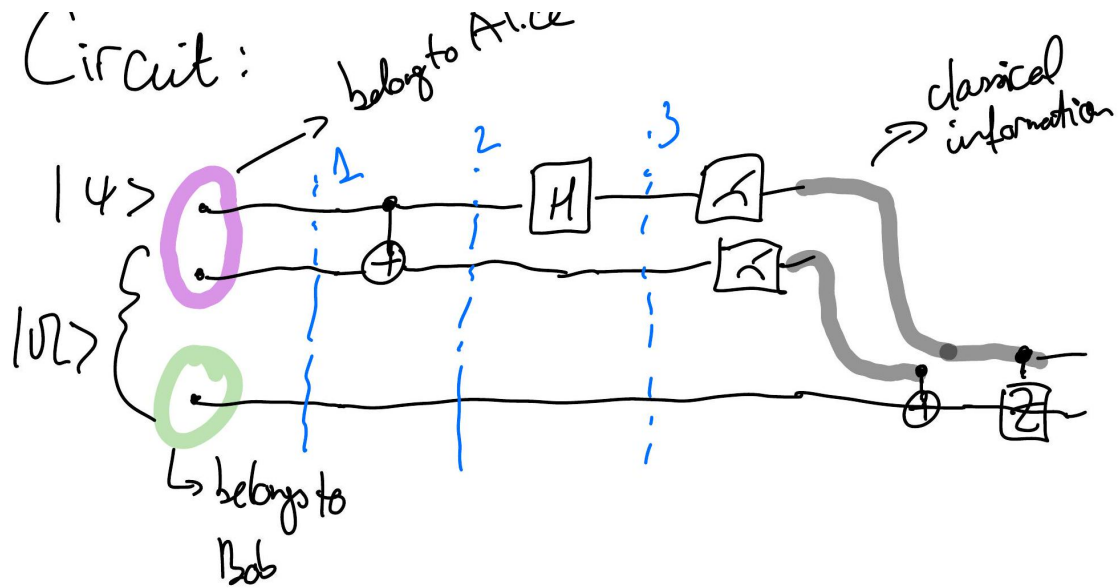


Figure 1.4: the circuit for quantum teleportation.

Analyzing the Protocol

- Suppose Alice measures $|00\rangle$. The probability is $\frac{1}{4}$. Then, Bob's state is $\alpha|0\rangle + \beta|1\rangle$ which is equivalent to $|\psi\rangle$. So Alice can tell him the outcome and he does nothing. He then knows he has $|\psi\rangle$ on his side.
- If Alice measures $|01\rangle$ or $|10\rangle$, Bob can adjust his state using appropriate quantum gates to recover $|\psi\rangle$. For instance, if Alice measures $|01\rangle$, Bob can apply X to recover $|\psi\rangle$. If it was $|10\rangle$, Bob can apply Z to recover $|\psi\rangle$.
- We conclude that the protocol works as it should.

This was one of the first protocols to showcase the power of quantum communication and how it is fundamentally different from classical communication!

1.1.16 Quantum Queries

We will now discuss how we will be able to access functions in a quantum computers. To do that, we will define various forms of quantum queries.

Classical queries:

Given $f : \{0, 1\}^n \rightarrow \{0, 1\}$, determine if f has some property. We have access to queries to the black-box implementing f . That is, we can ask for one x and obtain $f(x)$ in return. One example of a property could be e.g. whether f is constant. We then count the number of queries to the black-box. Note that we do not care how difficult it is to implement this black-box, we can ask it questions for free! Though this is a bit of a toy model, here we can easily demonstrate quantum speed-ups. But first we will define what quantum queries are.

Quantum queries:

model the query to f as a unitary operation:

$$O_f : \mathbb{C}^{2^n} \otimes \mathbb{C}^2 \rightarrow \mathbb{C}^{2^n} \otimes \mathbb{C}^2$$

They're defined as:

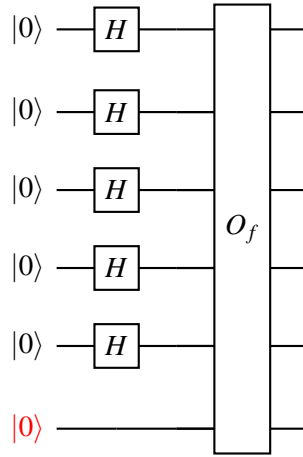


Figure 1.5: The circuit above prepares a uniform superposition of all queries. The address qubit is the one in red, and the circuit above first prepares the state $\frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x, 0\rangle$. Then, we can query this state to obtain $\frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x, f(x)\rangle$. This is a very powerful tool, and we will see how it can be used to obtain quantum speed-ups.

$$O_f : |x, b\rangle \rightarrow |x, f(x) \oplus b\rangle$$

Here, $\oplus$ denotes bitwise XOR. This operation allows for querying in superpositions, which is more powerful than classical queries. Note that we only need to specify it in a basis, since it is linear.

Note that some authors are of the opinion that comparing quantum and classical query complexity is not a fair comparison. This is because quantum queries are more powerful and it might be difficult to implement a quantum query. But one could counter that if we are able to implement a classical circuit that computes f , then we can also implement a quantum circuit that computes f . However, we will not worry about this for now.

2. Lecture 2

2.1 Quantum Circuits

Computation consists of a sequence of elementary operations.

Classically: An operation $g : \{0, 1\}^n \rightarrow \{0, 1\}^n$ that acts only on a few bits. For example, we might use the NOT operation on a single bit. These correspond to various gates in the device.

Quantumly: Tensor product of a unitary operator acting on a small number of qubits. For example, a Hadamard gate.

Notation: For an ordered set A of qubits in $\{1, \dots, n\}$ with h elements, and U being a unitary on h qubits, we define $U[A]$ as the unitary on A and identity everywhere else. For instance, $H[\{1\}]$ is represented as:

$$H \otimes \underbrace{I \otimes I \dots \otimes I}_{n-1}.$$

R $U[A]$ is a $2^n \times 2^n$ unitary matrix. While it is well-structured, multiplying several together to form $U_1[A_1]U_2[A_2] \dots U_m[A_m]$ can be complicated. Eventually, this approach becomes hard to simulate in polynomial time on a classical device.

Definition 2.1.1 — Quantum Circuit. Let $g \in U(2^h)$ be a fixed set of unitary operators, and each $u \in g$ acts on $(\mathbb{C}^2)^{\otimes h}$. A quantum circuit on n qubits with gate set g is a sequence $U_1[A_1], \dots, U_m[A_m]$, where A_i are ordered subsets of $\{1, \dots, n\}$ of size at most h , and $U_i \in g$.

• The operator defined by the circuit is given by $U = U_m[A_m] \dots U_1[A_1]$.

R g is called a gate set. It is usually taken to be finite.

R m is the size of the circuit.

- R** Gates acting on disjoint qubits can be applied in parallel. A circuit has depth $\leq d$ if it can be arranged in d layers of non-tumultuous gates.

The definition above does not consider the use of auxiliary systems, which are often crucial.

Definition 2.1.2 Given a circuit on N qubits, the unitary operator U on n qubits is realized by the unitary operator W with $N - n$ auxiliary systems if for all states $|\psi\rangle$:

$$(U|\psi\rangle)|0\rangle^{N-n} = W|\psi\rangle|0\rangle^{N-n}$$

It is important that in the definition above the auxiliary systems are initialized to $|0\rangle$ and that they are not entangled with the rest of the system. They can be understood as some form of catalyst.

2.1.1 Classical Computation with Quantum Circuits

It is not completely obvious at first that quantum computing can be used to simulate classical computing efficiently. That is, that for every classical circuit can be efficiently mapped to a quantum circuit that will compute the same function. We will now show that this is indeed the case. One of the main reasons for this difficulty is that the circuit model of quantum computing is *reversible*. That is, we can always recover the input state after applying a unitary U by applying $U^\dagger$. On the other hand, typical gate sets for classical computing are not reversible. For instance, the OR operation is not reversible, since we cannot distinguish if the inputs were $(0, 1)$ or $(1, 0)$ from the output, which is 1 in both cases.

Classical circuits comprise NOT, AND, and FANOUT (copy) gates, the sense that any Boolean function can be computed by using such gates. Thus, if we can show we can implement these gates on a quantum computer, then we are done.

One particularly simple way to do this is to resort to the TOFFOLI gate.



Figure 2.1: Tommaso Toffoli, who invented the gate in 1980 to come up with a reversible version of classical computation.

The action of the TOFFOLI gate (also known as CCNOT) is described in Fig. 2.2.

It defines a permutation on $(\mathbb{C}^2)^{\otimes 3}$ and is unitary. Thus, it is in particular reversible and implementable on a quantum computer.

Let us now show how to implement a universal set of gates with the Toffoli only. We will start by showing how to implement the AND gate in Fig. 2.3

The NOT gate is represented in Fig. 2.4.

The FANOUT gate is constructed in Fig. 2.5.

Conclusion: We can transform an m -gate classical circuit consisting of those gates into a quantum circuit consisting of Toffoli gates! The number of qubits in the circuit $\leq 3m$. How do we do it? For each gate in the original circuit, we potentially append 2 qubits and get the original gate

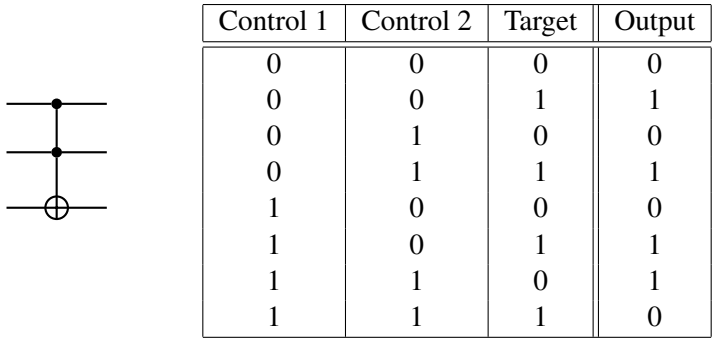


Figure 2.2: The circuit diagram and the truth table for the TOFFOLI gate. Note that it defines a permutation on 3 bits, which means that it can also define a unitary matrix on 3 qubits by just permuting the computational basis.

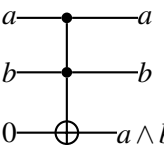


Figure 2.3: Representing a classical AND gate with the Toffoli gate.

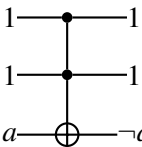


Figure 2.4: Obtaining a classical NOT gate with the Toffoli gate.

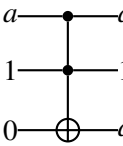


Figure 2.5: Representing a classical FANOUT gate with the Toffoli gate.

by applying a Toffoli gate on these three qubits instead. Repeating this for every gate gives us the desired result. But note that this will give us some garbage qubits we will not use at the end of the computation. Note that also the algorithm to compile the new circuit with Toffolis runs in linear time in the number of gates.

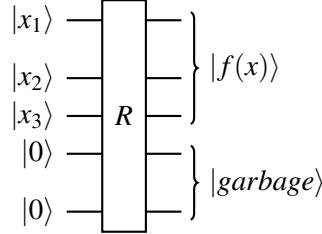


Figure 2.6: How the reversible circuit looks like. We have the input qubits and some auxiliary ones that will be used to convert the irreversible gates into reversible ones with Toffolis. The output in the first qubits is $f(x)$ and we get some extra garbage.

It may be desirable to "clean" the garbage in a systematic manner. This can be done by applying the same circuit in reverse. This is shown in Fig. 2.7. Note that the circuit in Fig. 2.7 implements the oracle discussed earlier, implying that if we have an efficient circuit to implement a function, we also have an efficient quantum circuit that implements the quantum oracle corresponding to that function efficiently. This justifies the quantum oracle model as being a reasonable model of computation.

2.2 The Early Quantum Algorithms

The first quantum speed-ups were discovered in the context of query complexity.

Queries: Given $f : \{0, 1\}^n \rightarrow \{0, 1\}$, we want to determine if f has some property. We count the number of queries to the black-box. This is admittedly a bit of a toy model, but here, we can easily prove quantum speed-ups. For quantum queries, we model a query to f as unitary, $O_f : \mathbb{C}^{2^n} \otimes \mathbb{C}^2 \rightarrow \mathbb{C}^{2^n} \otimes \mathbb{C}^2$.

It is defined as $O_f : |x, b\rangle \rightarrow |x, f(x) \oplus b\rangle$, which is a bitwise operation/xor, and it's extended linearly.

Quantum queries are more powerful than classical queries because we can perform queries in superposition!

■ **Example 2.1 — Uniform superposition.** Consider the circuit in Fig. 1.5:

We can show that

$$|\psi_1\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x, 0\rangle$$

and then

$$|\psi_2\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x, f(x)\rangle.$$

■

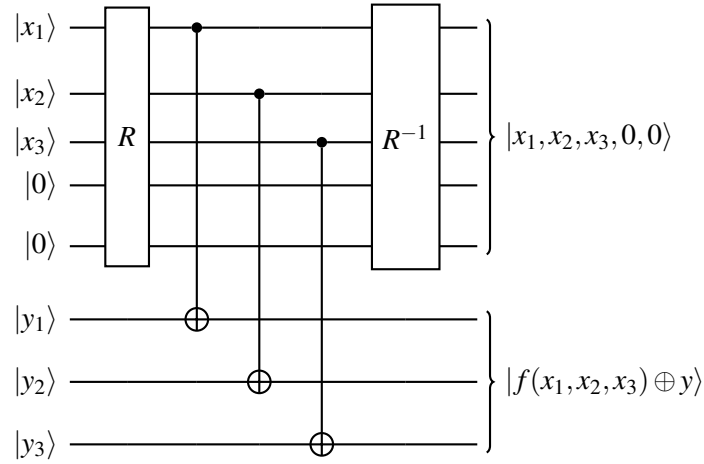


Figure 2.7: Cleaning the garbage. By implementing the reversible circuit this way, we can clean the garbage in a systematic manner and put the result of the function $f(x) \in \{0, 1\}^m$ of the original circuit in the last m qubits. Thus, we can implement quantum oracles

What do we obtain as an outcome when measuring $|\psi_2\rangle$? Is this useful? Important fact: we just used that

$$H^{\otimes n}|0^n\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle.$$

More generally, we have:

$$H^{\otimes n}|i\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} (-1)^{i \cdot x} |x\rangle$$

where $i \cdot x$ means $\sum_{j=0}^{n-1} i_j \cdot x_j$.

■ **Example 2.2 — Phase oracle.** The oracle we defined embeds the information about $f(x)$ in the state of the second register. However, we can also encode the information in the phase.

Note that:

$$O_f|x\rangle \left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle \right) = \frac{1}{\sqrt{2}}|x, f(x)\rangle - \frac{1}{\sqrt{2}}|x, f(x) \oplus 1\rangle.$$

We will usually refer to the state $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ as the *minus state*.

- If $f(x) = 0$, we obtain:

$$\frac{1}{\sqrt{2}}(|x, 0\rangle - |x, 1\rangle) = |x\rangle|-\rangle.$$

- If $f(x) = 1$, we obtain:

$$-\frac{1}{\sqrt{2}}|x, 0\rangle + \frac{1}{\sqrt{2}}|x, 1\rangle = -|x\rangle|-\rangle.$$



Figure 2.8: David Deutsch and Richard Jozsa.

In general:

$$O_f|x\rangle|-\rangle = (-1)^{f(x)}|x\rangle|-\rangle.$$

■

2.3 The Deutsch-Jozsa Algorithm

This algorithm was developed by David Deutsch and Richard Jozsa (1998). It gives the first separation between quantum query complexity and deterministic classical query complexity.

Problem Setting: We are given

$f : \{0, 1\}^n \rightarrow \{0, 1\}$ with the promise that it's either balanced or constant. If f is constant, then we have:

$$f(x) = 1 \text{ or } 0 \quad \forall x \in \{0, 1\}^n.$$

If f is balanced, then for half of the strings we have $f(x) = 0$ and for the other half $f(x) = 1$. Our task is to determine which type of function f is with as few queries to f as possible.

Question: How many queries to f are required classically for a deterministic algorithm?

Answer: $2^{n-1} + 1$. Quantumly?

The quantum circuit for the Deutsch-Jozsa algorithm is shown in Fig. 2.9. Let us analyse it step-by-step.

Step 1: We will compute the state after applying O_f . This corresponds to the output of the phase oracle:

$$|\psi_1\rangle = \frac{1}{\sqrt{2^n}} \sum_x (-1)^{f(x)} |x\rangle |-\rangle.$$

Step 2: We will now compute the state after applying the second layer of Hadamards. Recall that

$$H^{\otimes n}|x\rangle = \sum_j \frac{(-1)^{x \cdot j}}{\sqrt{2^n}} |j\rangle.$$

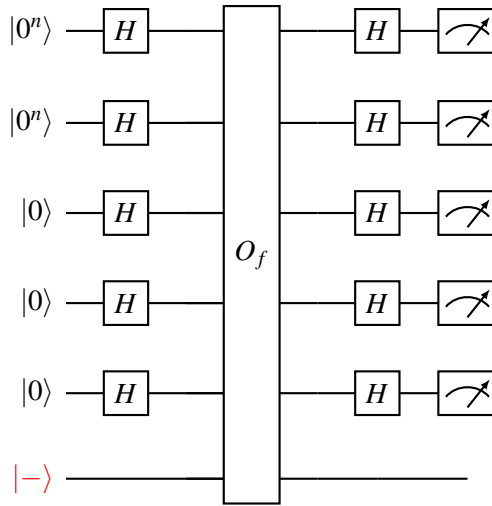


Figure 2.9: the quantum circuit for the Deutsch-Josza algorithm, here on $n = 5$ qubits. Note that the address qubit is initialized in the minus state, which means we are implementing the phase oracle.

We then have:

$$|\psi_2\rangle = \frac{1}{2^n} \sum_x (-1)^{f(x)} \sum_j (-1)^{x \cdot j} |j\rangle |- \rangle$$

Let's compute the probability of observing $|0^n\rangle$ when measuring $|\psi_2\rangle$.
Setting $j = 0$:

$$\sum_x (-1)^{f(x)} |0\rangle \rightarrow \begin{cases} 2^n & \text{if it is constant} \\ 0 & \text{if it is balanced} \end{cases}$$

Thus, if we observe $|0^n\rangle$ when measuring, the function is constant. If it's any other string, it's balanced. Quantumly, we only need 1 query. This gives us a classical requirement of $2^{n-1} + 1$ versus just 1 query quantumly.

However, our comparisons thus far have been exclusively with deterministic algorithms, without accounting for algorithms that allow a small probability of failure. Now, let's delve into the Bernstein-Vazirani algorithm:

Bernstein-Vazirani Algorithm

Proposed by Umesh Vazirani and Ethan Bernstein in 1997. It is a generalization of the Deutsch-Josza algorithm.

Problem Setting: Given a function f such that $f(x) = x \cdot a \bmod 2$, where $a \in \{0, 1\}^n$, our objective is to determine the vector a .

Classically: Each query (deterministic or random) reveals one bit of information about a . Thus, to completely determine a , we need at least n queries, or $\Omega(n)$ in big O notation.

Quantumly: We can employ a similar circuit as used for the Deutsch-Josza algorithm. When querying with y , we get:

$$|\psi_1\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} (-1)^{a \cdot x} |x\rangle.$$

But now notice that $H^2 = I$ and $H^{\otimes n} |a\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} (-1)^{a \cdot x} |x\rangle \Rightarrow H^{\otimes n} |\psi_1\rangle = |a\rangle$.

Thus, the circuit of Deutsch-Josza will give a ! We only need 1 query quantumly and $\Omega(n)$ for classical algorithms.

Bibliography

Books

Articles

Index

C

Contex7